UPLOAD SLOT 10 OF 12 · COMPLETE DESIGN REPORT

Complete Design Report

Falaj Al Safa — concept and preliminary design for Al Safa 2 Park

A 15,000 m² neighbourhood park in Dubai, redesigned around one question: how do you make an outdoor space usable in a city that is too hot to stand in? The answer proposed here raises comfortable daylight hours from 44.5% to 64.6%, within a capital cost of AED 27.0 M against the AED 35 M budget.

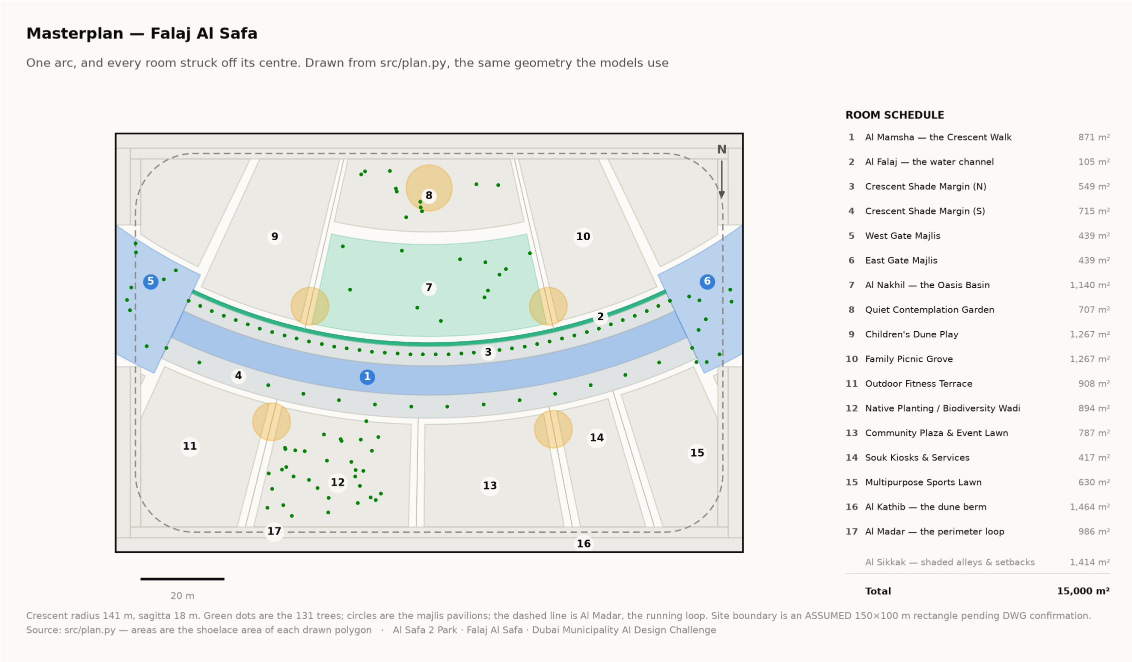
Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai · Mohamed Wasim, Individual Applicant · Issued 03 August 2026 · every quantity regenerated by `python run_analysis.py`

1. The scheme

One arc. A crescent of shade 141 m in radius sweeps across the site, bowing convex south so that its hollow opens to the north. A 0.9 m water channel runs beneath its northern drip line. Every room in the park is struck off the same centre, so no room is a rectangle and every room faces the crescent square-on.

Element	What it is	Description
Al Hilal	the Crescent Canopy	18 m gridshell at 4.5 m over a 7 m walk, with a 3 m southern louvre
Al Falaj	the water channel	0.9 m wide, on the canopy's drip line so it is shaded all day and evaporates less
Al Nakhil	the Oasis Basin	a sunken palm court held in the crescent's concave side
Al Sikkak	the alleys	radial — each one a radius of the same arc
Al Madar	the perimeter loop	a tree-shaded running and walking circuit
Al Kathib	the dune berm	planted earth against the roads — noise, glare and heat

The six named elements. All are generated from a single arc definition in src/plan.py.



Masterplan. Technical drawing — to scale.

2. Performance

Measure	Today	As designed
Comfortable daylight hours	44.5%	**64.6%**
Peak heat index	56.8 °C	**48.7 °C**
Crescent Walk shaded	—	**87.3%**
Site-wide mean shade	—	34.1%
Trees	—	131

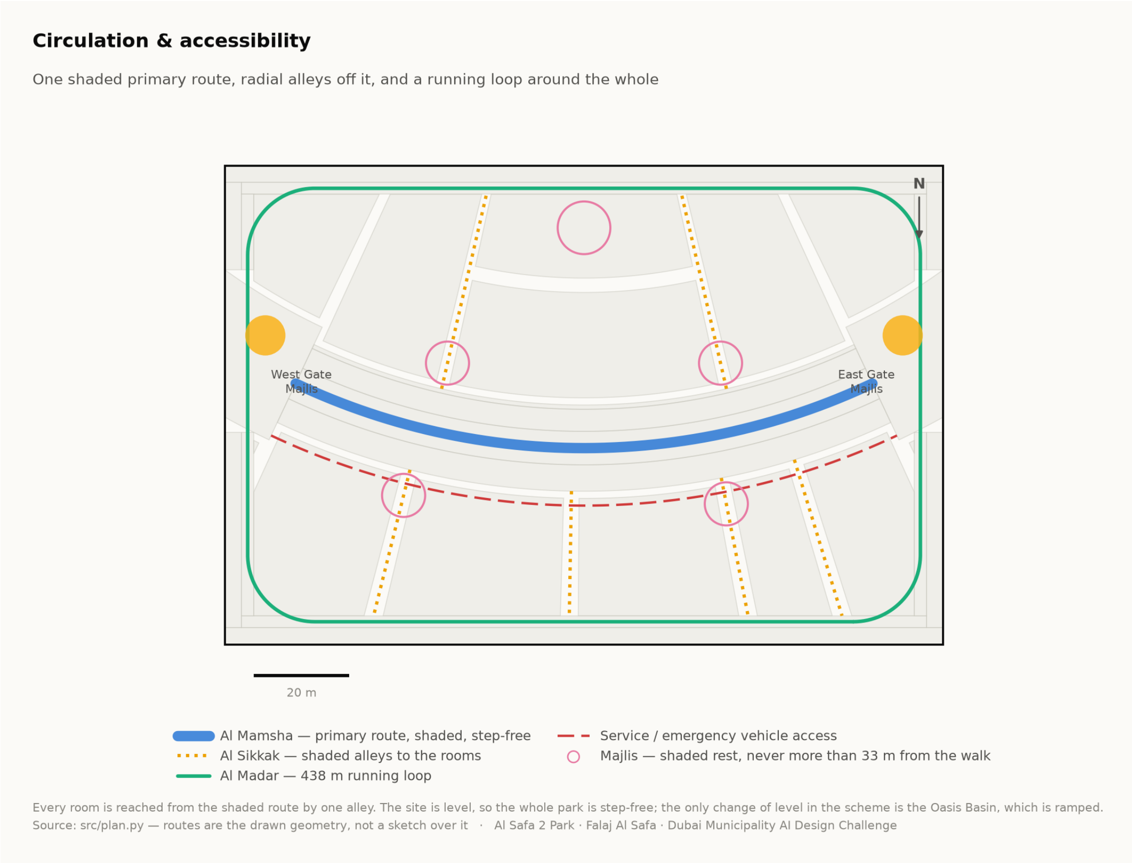
3. Room schedule

Room	Category	Area (m²)	% of site
Al Mamsha — the Crescent Walk	Circulation	871	5.81%
Al Falaj — the water channel	Water	105	0.70%
Crescent Shade Margin (N)	Green	549	3.66%
Crescent Shade Margin (S)	Green	715	4.77%
West Gate Majlis	Arrival	439	2.93%
East Gate Majlis	Arrival	439	2.93%
Al Nakhil — the Oasis Basin	Green	1,140	7.60%
Quiet Contemplation Garden	Passive	708	4.72%
Children's Dune Play	Active	1,267	8.45%
Family Picnic Grove	Passive	1,267	8.45%
Outdoor Fitness Terrace	Active	908	6.06%
Native Planting / Biodiversity Wadi	Green	894	5.96%
Community Plaza & Event Lawn	Social	788	5.25%

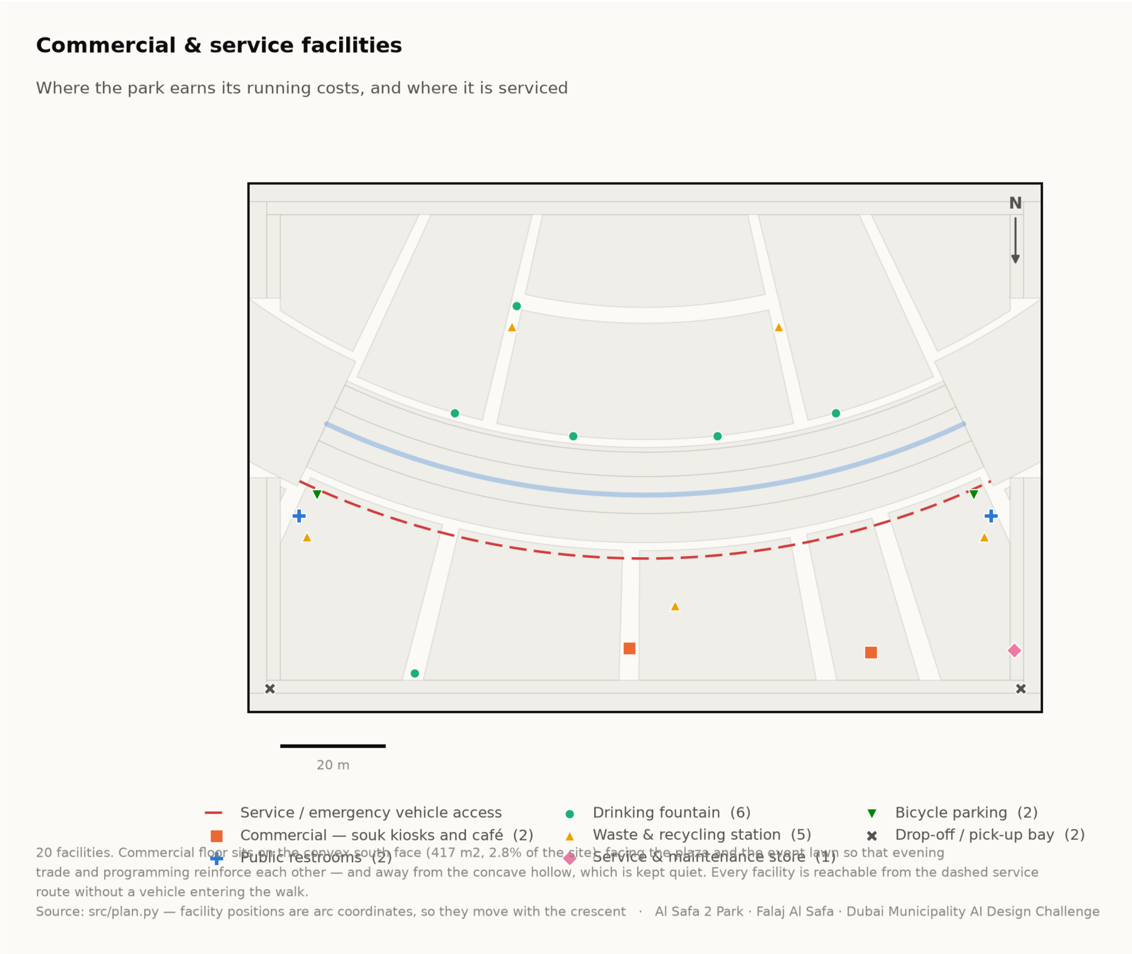
Room	Category	Area (m²)	% of site
Souk Kiosks & Services	Commercial	417	2.78%
Multipurpose Sports Lawn	Active	630	4.20%
Al Kathib — the dune berm	Green_Buffer	1,464	9.76%
Al Madar — the perimeter loop	Circulation	986	6.57%
Al Sikkak — shaded alleys & setbacks	Circulation	1,414	9.43%

Total 15,000 m².

4. Circulation, accessibility and facilities



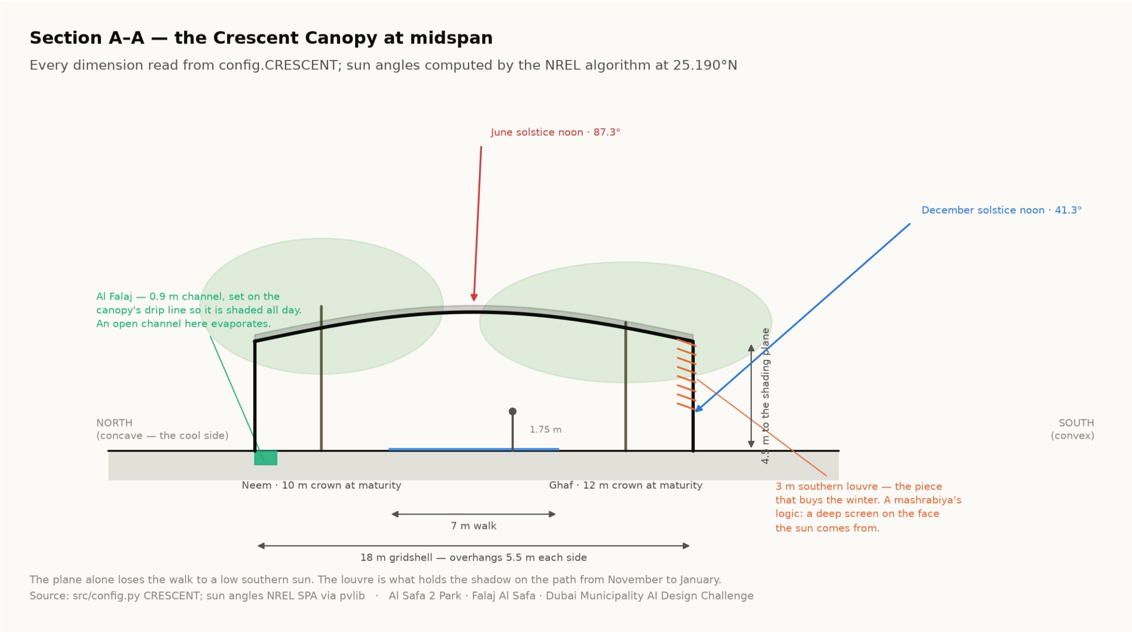
Circulation and accessibility. Technical drawing — to scale.



Commercial and service facilities, with the service route. Technical drawing — to scale.

20 facilities are placed against the drawn geometry: 2 commercial, 2 restroom blocks, 6 drinking fountains, 5 waste and recycling points, 1 maintenance store, 2 bicycle parking areas and 2 drop-off bays. Each is reachable from the service route without a vehicle entering the walk.

5. The section that makes it work



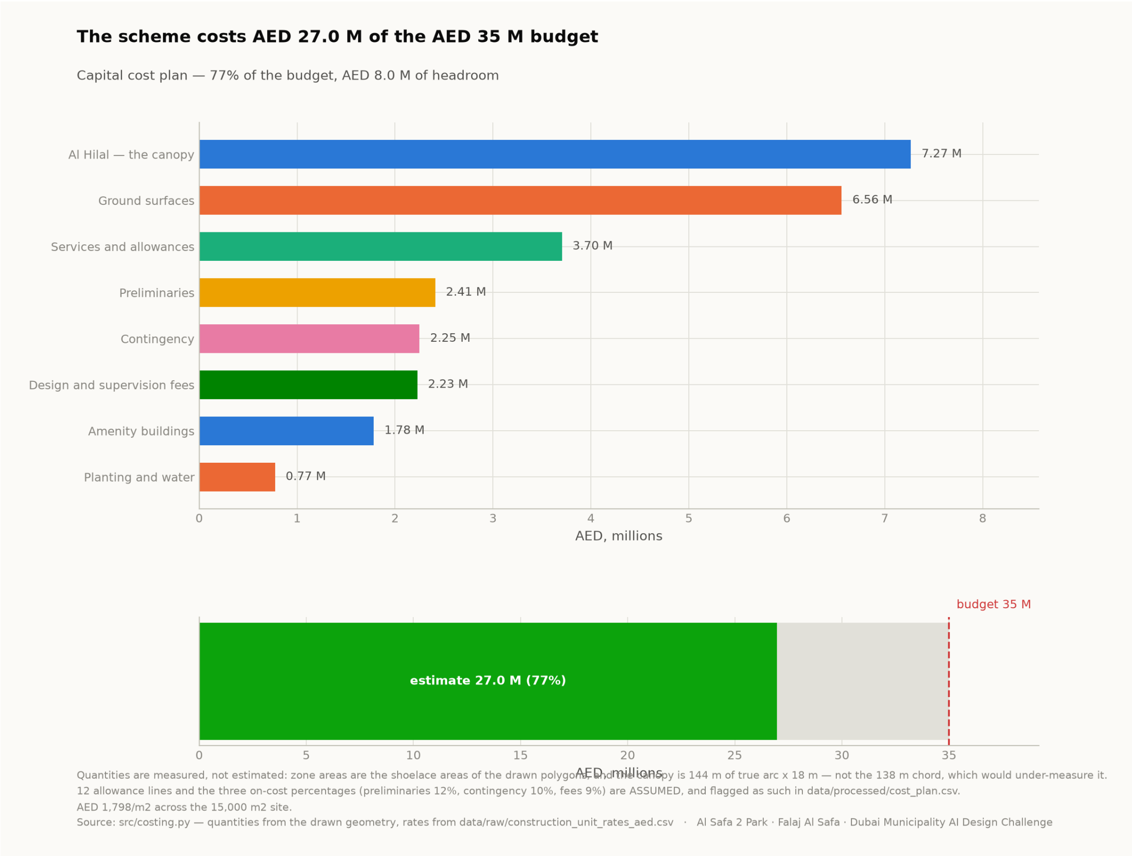
Section A–A: a 7 m walk under an 18 m gridshell at 4.5 m, with a 3 m southern louvre. Sun angles computed, not drawn. Technical drawing — to scale.

6. Feasibility — cost against the AED 35 M budget

The capital cost plan measures quantities from the drawn geometry rather than estimating them. Zone areas are shoelace areas; the canopy is priced on 144.2 m of *true arc* — not the 138 m chord, which would under-measure the structure by more than 6 m.

Element	AED millions
Al Hilal — the canopy	7.27
Ground surfaces	6.56
Services and allowances	3.70
Amenity buildings	1.78
Planting and water	0.77
Preliminaries	2.41
Contingency	2.25
Design and supervision fees	2.23
TOTAL	**26.97**
Budget	**35.00**
Headroom	**8.03**

Total AED 26,973,013, or 77.1% of the budget, at AED 1,798/m² across the site. Allowance lines and the three on-cost percentages are flagged ASSUMED in data/processed/cost_plan.csv.



Capital cost plan against the budget. Analysis output — computed from project data.

7. Honesty notes

- The **site boundary is assumed** — a 150 × 100 m rectangle pending confirmation against the issued CAD file. Every area figure depends on it.
- The **climate series is modelled** from 39 years of monthly normals, verified back to within 0.39 °C, and labelled as modelled everywhere it appears.
- **Visitor demand is a scenario**, not a prediction, and is deliberately excluded from the machine learning suite.
- **Diffuse radiation is excluded** from the shade model, so shaded comfort is stated conservatively.
- The **photovoltaic array runs a deficit**, covering about 13% of load — not a surplus.
- The **renders are illustrations**, not photographs of a built thing, and none is presented as analysis.

8. Corrections this submission makes to itself

An earlier version of this project claimed 99.2% annual shade. It did not survive a geometric check and was withdrawn; the re-solved section measures 87.3%. Three visuals that presented invented data as measurement were also withdrawn. Both corrections are stated here rather than quietly removed, because a submission that audits itself in public is more credible than one that never had to.